HBESC vs Gaussian Fill: Key Figures and Trajectories

Compact Phase 6 PDF report with selected aggregate figures and representative trajectories. The report is intentionally bounded to key evidence so it stays under the requested 20-page ceiling.

Technical Summary

The visual evidence supports a narrow story: baseline HBESC works well on tested single-well quadratic/quartic landscapes, conservative one-fill Gaussian fill did not rescue the quartic double-well failure mode, and physical feasibility is a first-order result rather than a footnote.

84 Phase 5B execute rows, all sim-time complete	**14** Rows entered the 2 m success radius	**0** Double-well expanded rows entered the target radius	**11 / 14** Phase 5B success entries exceeded TurtleBot3 reference limits

How To Read This Report

Figures 1-3 give the aggregate picture. Figures 4-11 show selected trajectories: the Class A quartic pair, the physical-limit tradeoff, gain sensitivity, the strongest Gaussian-fill pilot result, and the unresolved double-well failure pair.

Scope And Definitions

Success radius: first entry into the 2 m target radius. **Final distance:** distance to target at the end of the bounded run. **Physical label:** command-derived comparison against TurtleBot3 Burger reference limits, especially wheel RPM.

Scenario	Method	Final dist. m	Success time s	Physical label	Fill count
P5B-PILOT-02-G2_GLOBALW40_START1_REAL	baseline_hbesc	1.394	59.4	exceeds_turtlebot3_burger_reference_limits	0
P5B-PILOT-GF-G2_GLOBALW40_START1_REAL	gaussian_fill	0.159	284.4	within_turtlebot3_burger_reference_limits	1
P5B-CURV-DEFAULT-HB	baseline_hbesc	0.580	143.2	exceeds_turtlebot3_burger_reference_limits	0
P5B-CURV-DEFAULT-GF	gaussian_fill	5.695	not reached	within_turtlebot3_burger_reference_limits	1
P5B-DW-A0P01-REAL-HB	baseline_hbesc	11.314	not reached	exceeds_turtlebot3_burger_reference_limits	0
P5B-DW-A0P01-GF	gaussian_fill	13.338	not reached	within_turtlebot3_burger_reference_limits	1

Source metrics: `results/batches/phase5_expanded_execute/phase5_expanded_metrics.csv`, `phase5_expanded_summary.json`, Phase 3/4 result documents, and selected per-run trajectory PNGs.

Phase 5B success-rate heatmap

Success means first entry into the 2 m target radius; Phase 5B, 84 execute rows.

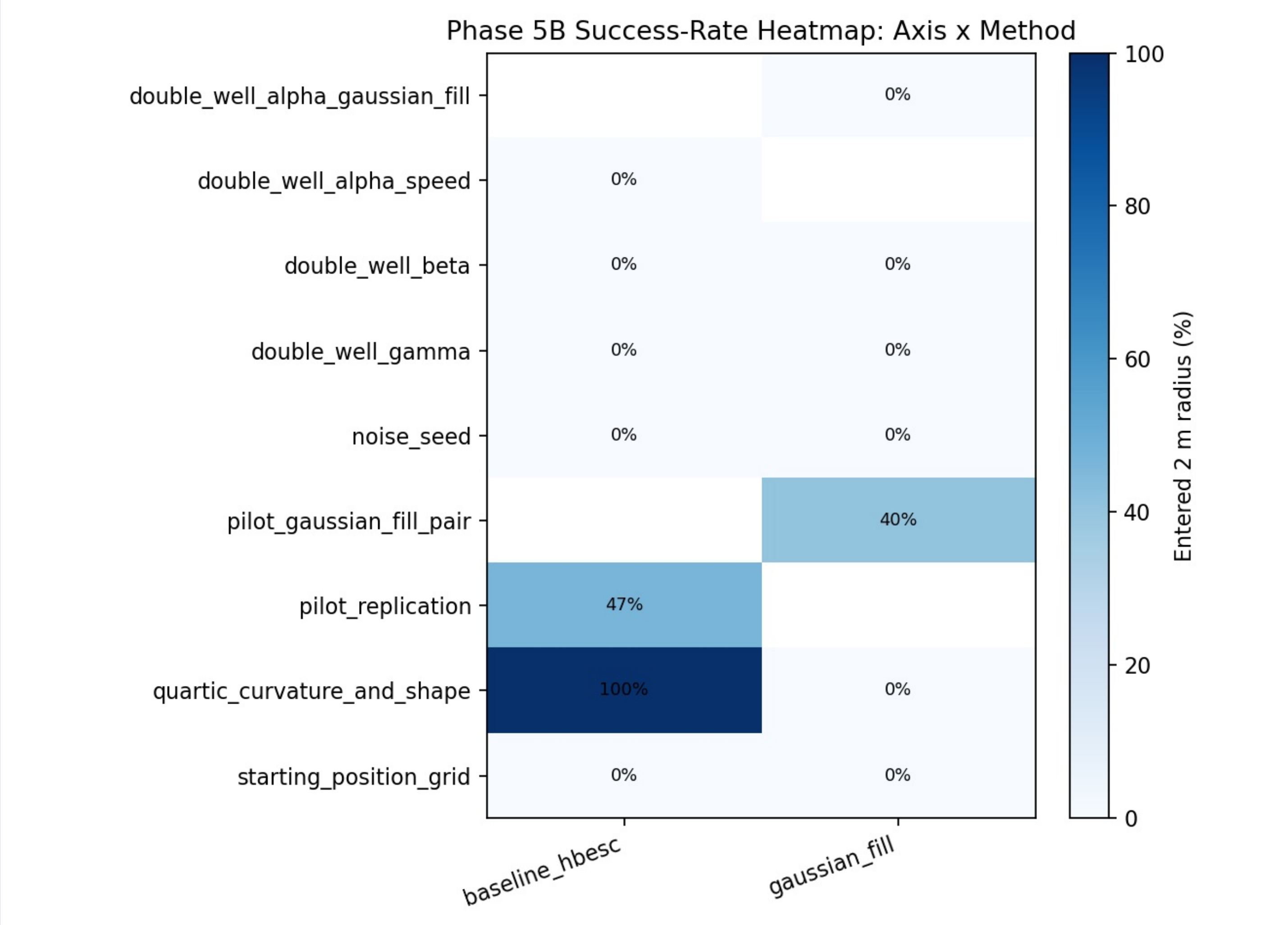

Figure 1. Baseline HBESC successes concentrate in pilot and single-well quartic rows; tested double-well slices did not enter the target radius.

Double-well alpha x speed heatmap

Final distance to target for baseline HBESC alpha-speed rows; lower is better.

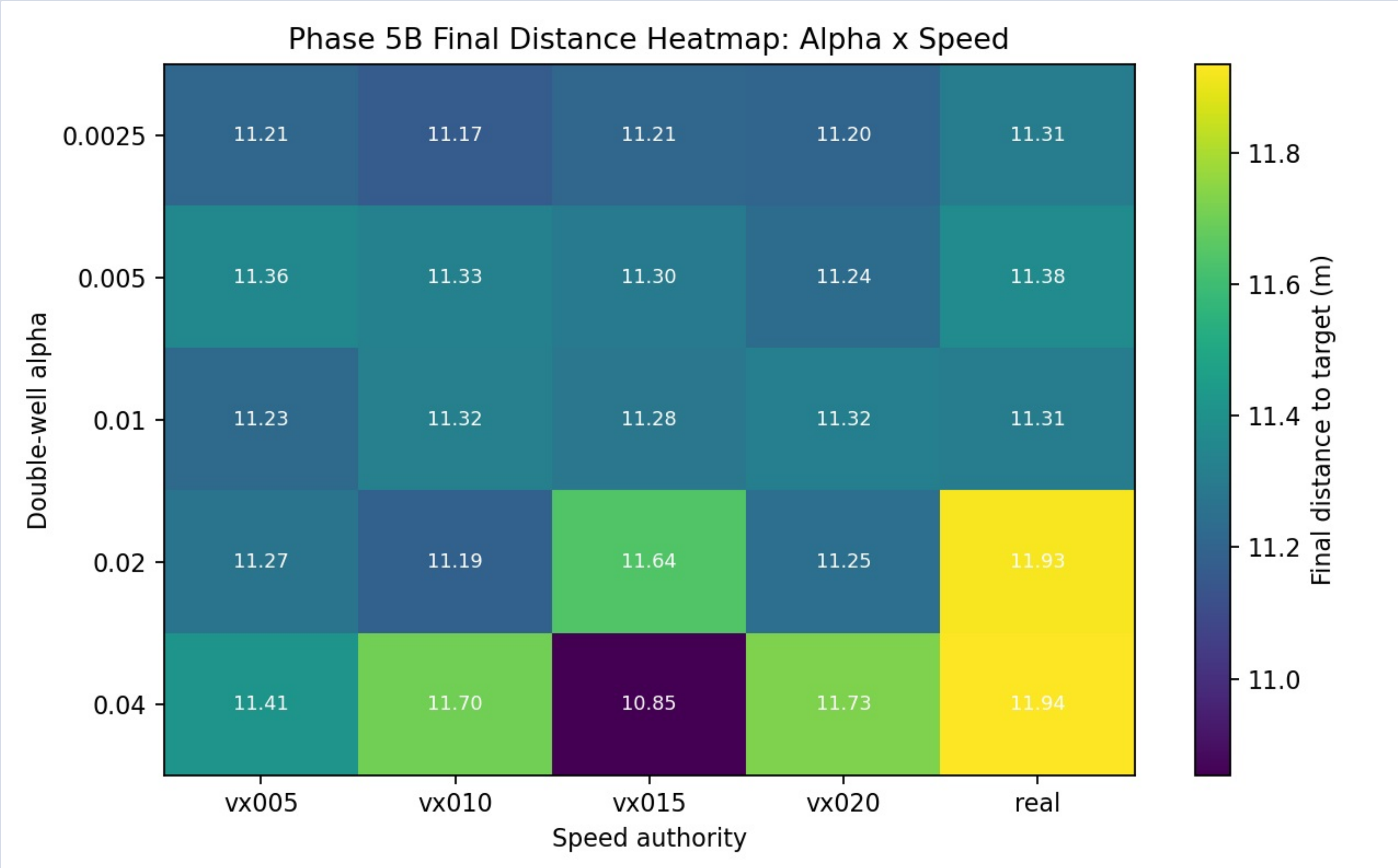

Figure 2. Increasing speed authority did not produce an escape boundary in the tested quartic double-well family.

Physical feasibility labels

Phase 5B rows labeled against TurtleBot3 Burger reference limits.

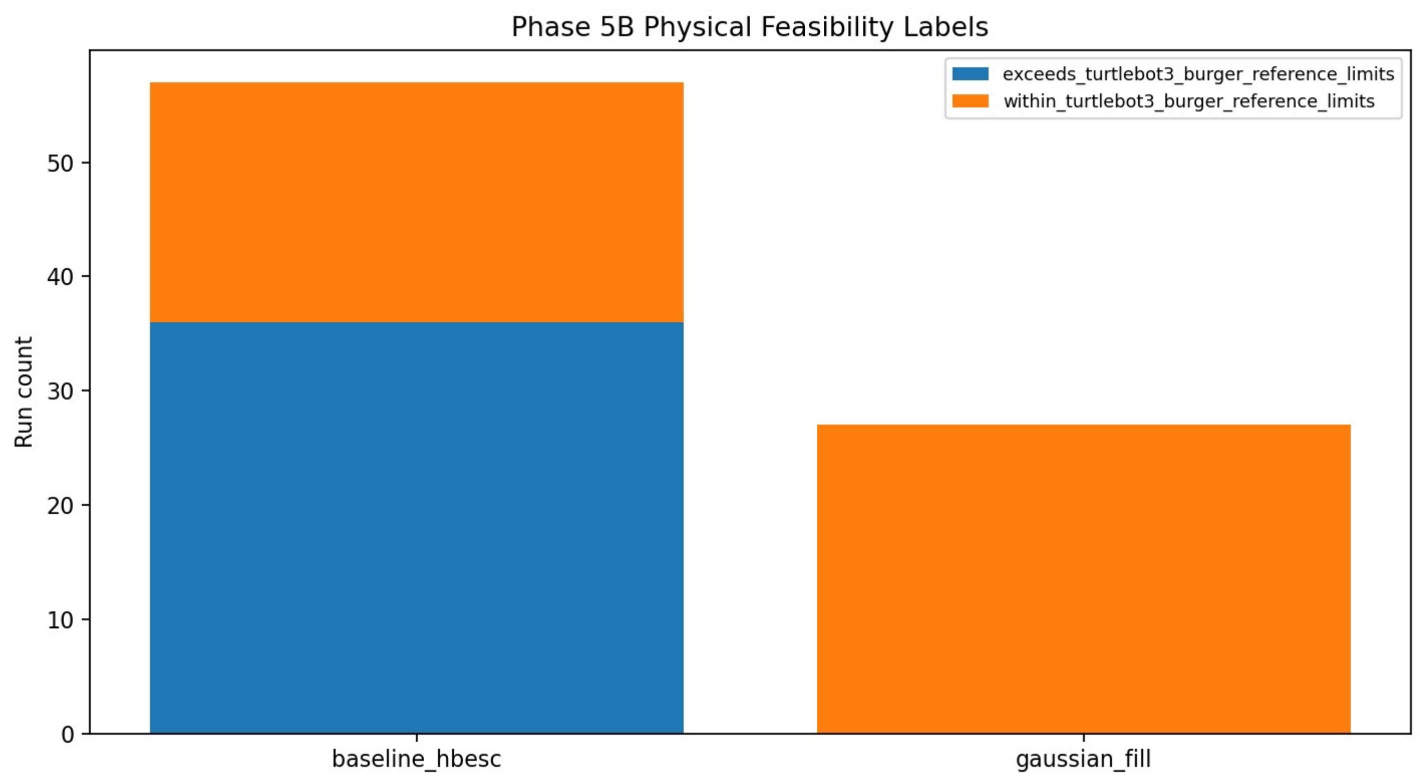

Figure 3. Gaussian-fill rows stayed within reference limits; many baseline rows, including most baseline successes, exceeded them.

Trajectory: convex quartic baseline HBESC success

QRT-A-HB, Phase 3; final distance 0.565 m.

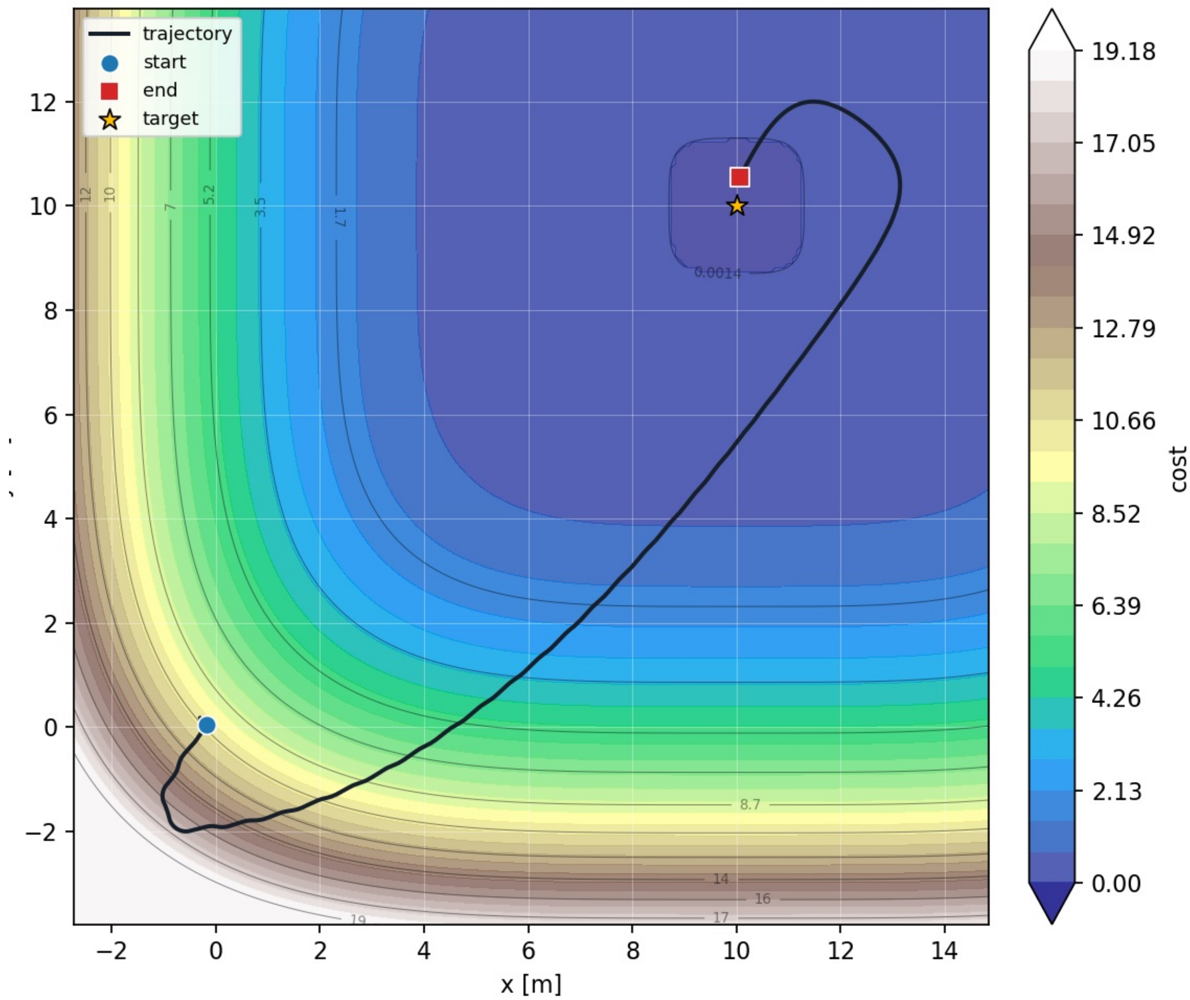

Figure 4. Baseline HBESC reaches the single-well quartic target region cleanly, supporting Class A evidence.

Trajectory: convex quartic Gaussian fill underperforms

QRT-A-GF, Phase 3; final distance 5.786 m, one fill.

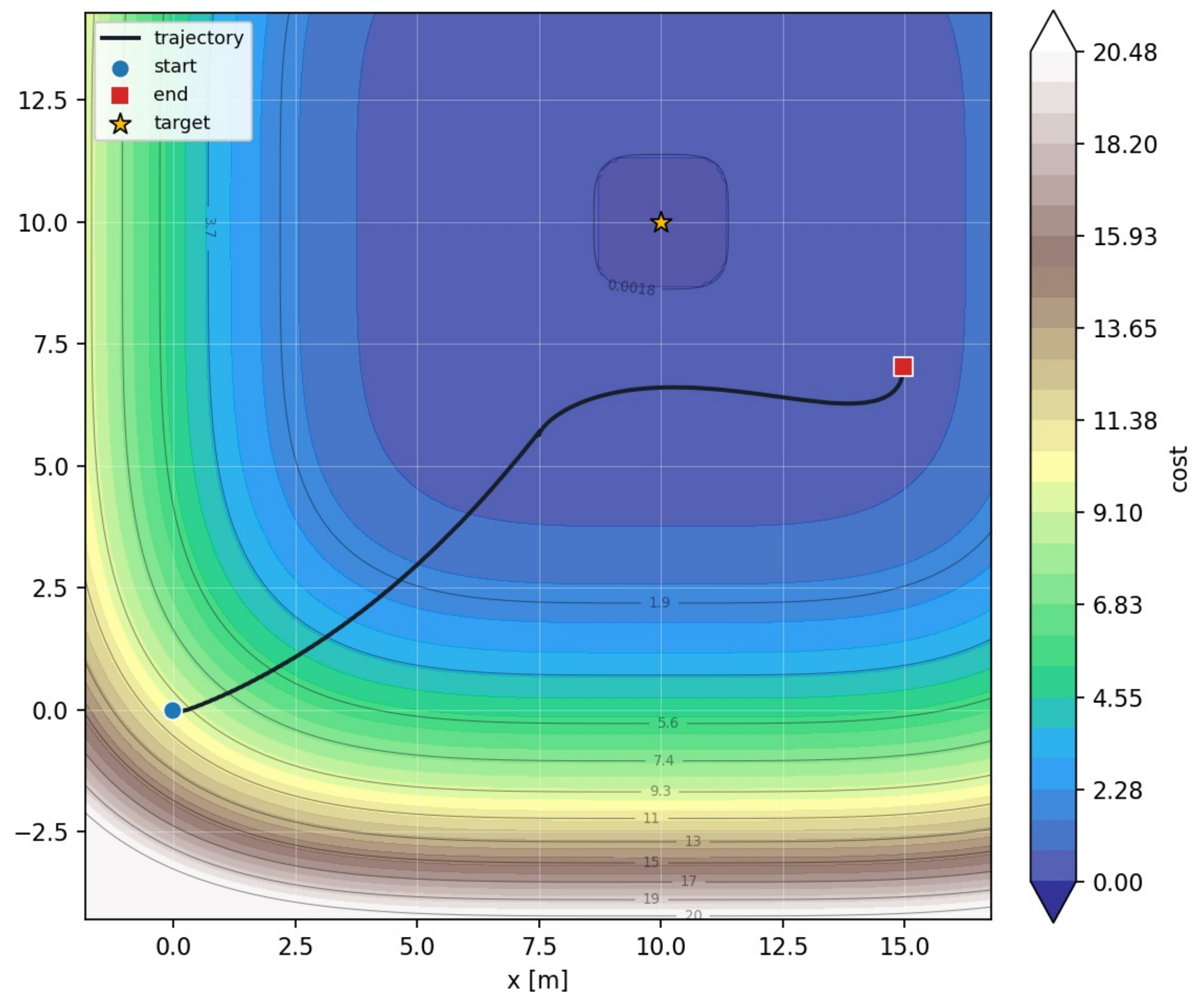

Figure 5. The paired Gaussian-fill run did not enter the 2 m target radius, showing that fill can distort an otherwise useful trajectory.

Trajectory: physically feasible slow baseline

QRT-C-SLOW, Phase 3; final distance 0.517 m, success time 267.308 s.

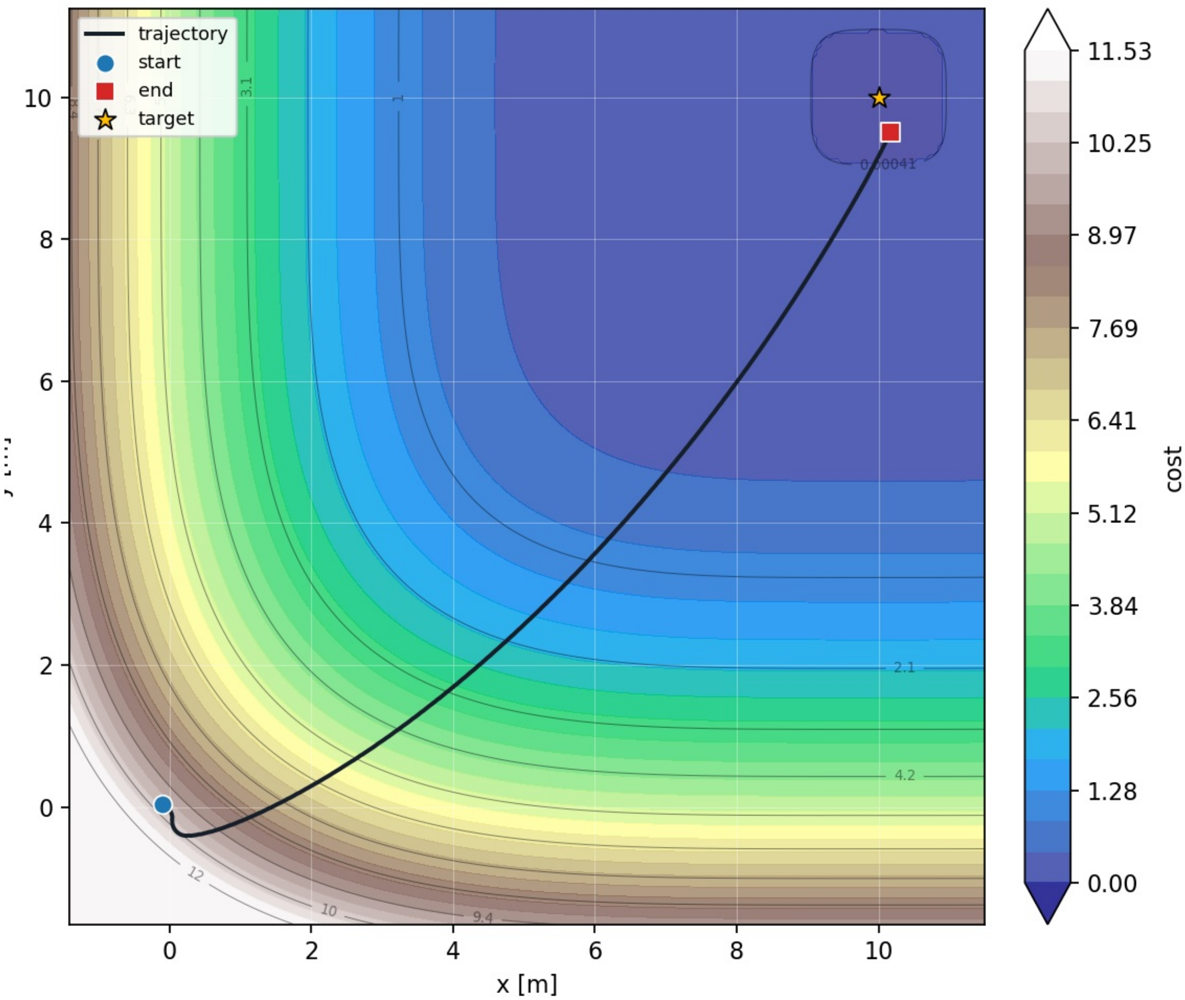

Figure 6. The low-speed baseline can converge while staying within reference limits, but it takes substantially longer.

Trajectory: real-speed baseline converges but exceeds limits

QRT-C-REAL, Phase 3; final distance 0.439 m, success time 144.670 s.

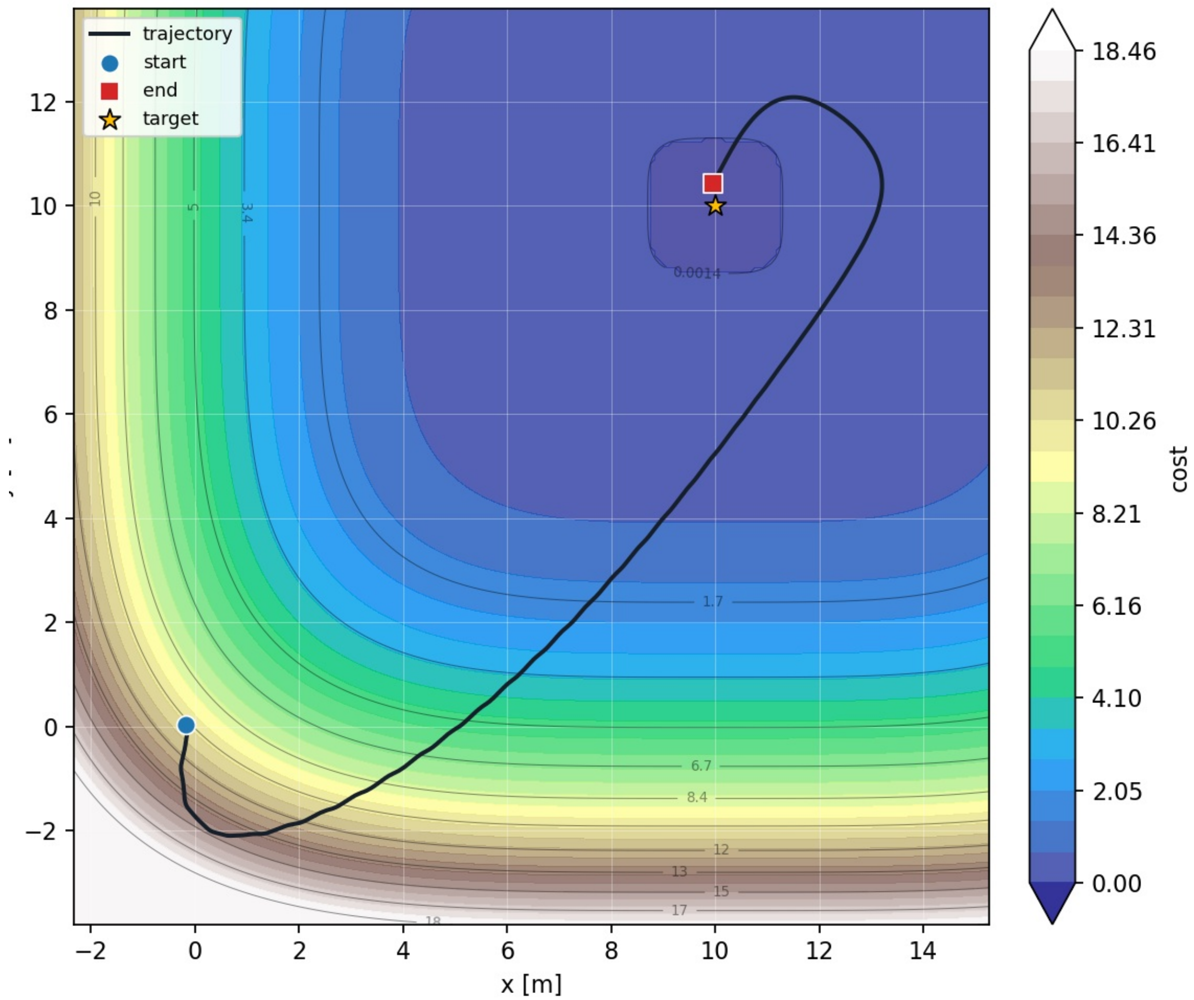

Figure 7. The faster real-speed run converges earlier but exceeds TurtleBot3 Burger reference wheel limits.

Trajectory: high HeavyBall k improves convex quartic convergence

QRT-GAIN-HBK-HIGH, Phase 4; final distance 0.400 m.

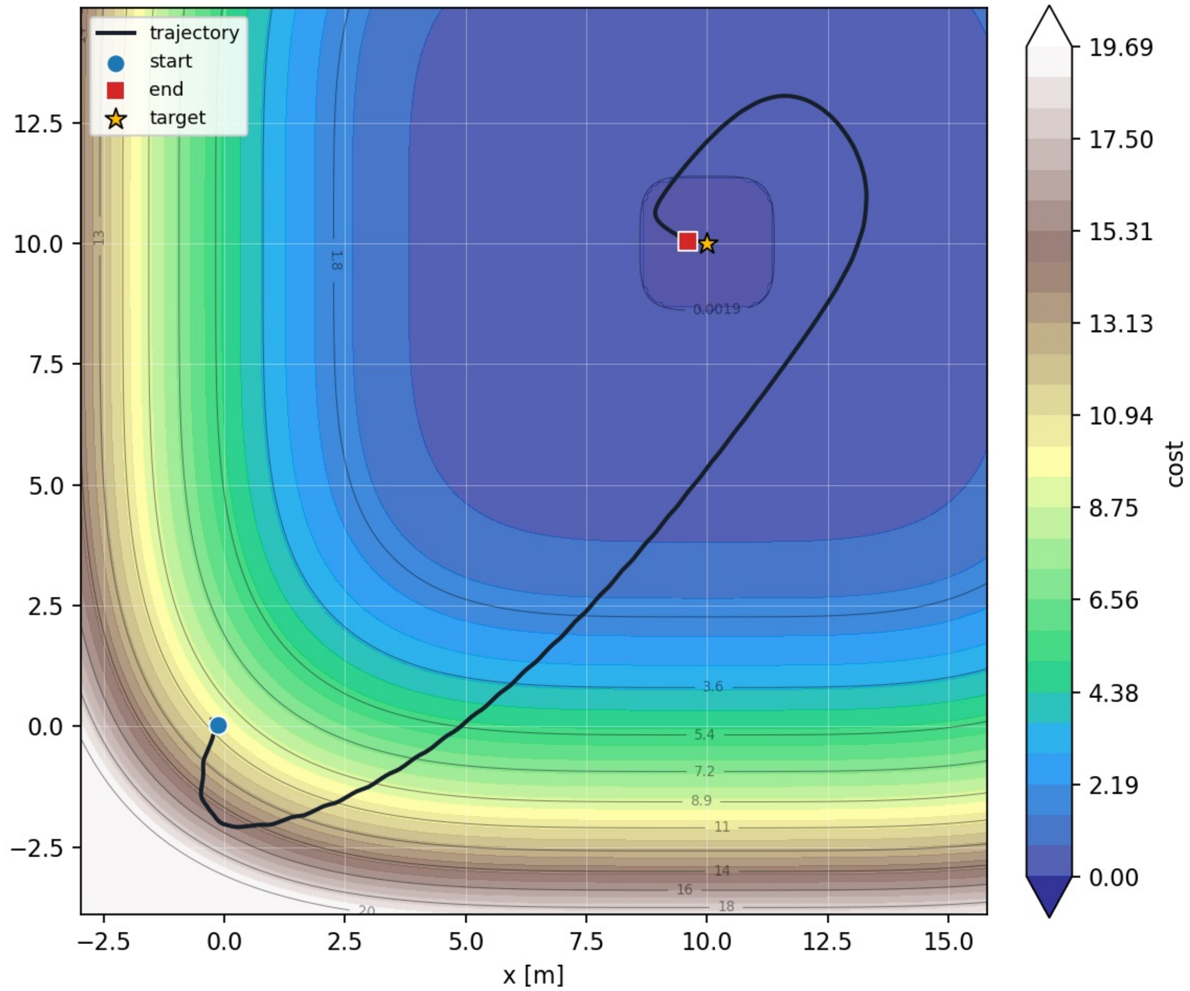

Figure 8. High HeavyBall k produced the best Phase 4 final distance, but it still exceeded TurtleBot3 Burger reference limits.

Trajectory: Gaussian fill two-basin pilot success

P5B-PILOT-GF-G2_GLOBALW40_START1_REAL; final distance 0.159 m.

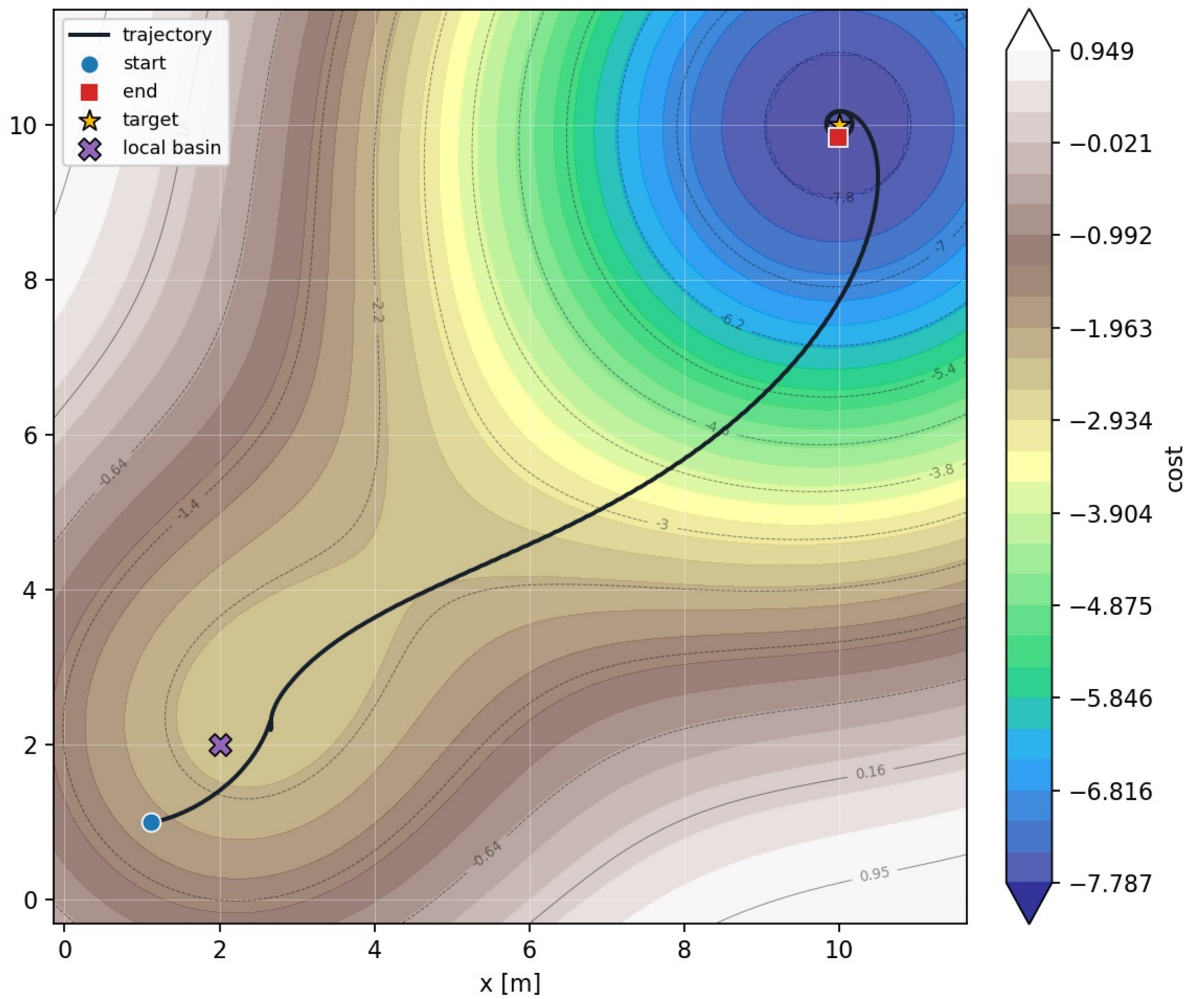

Figure 9. This is the strongest Gaussian-fill result: clean final tracking and within reference limits on a Gaussian two-basin case.

Trajectory: double-well baseline remains trapped

P5B-DW-A0P01-REAL-HB; final distance 11.314 m.

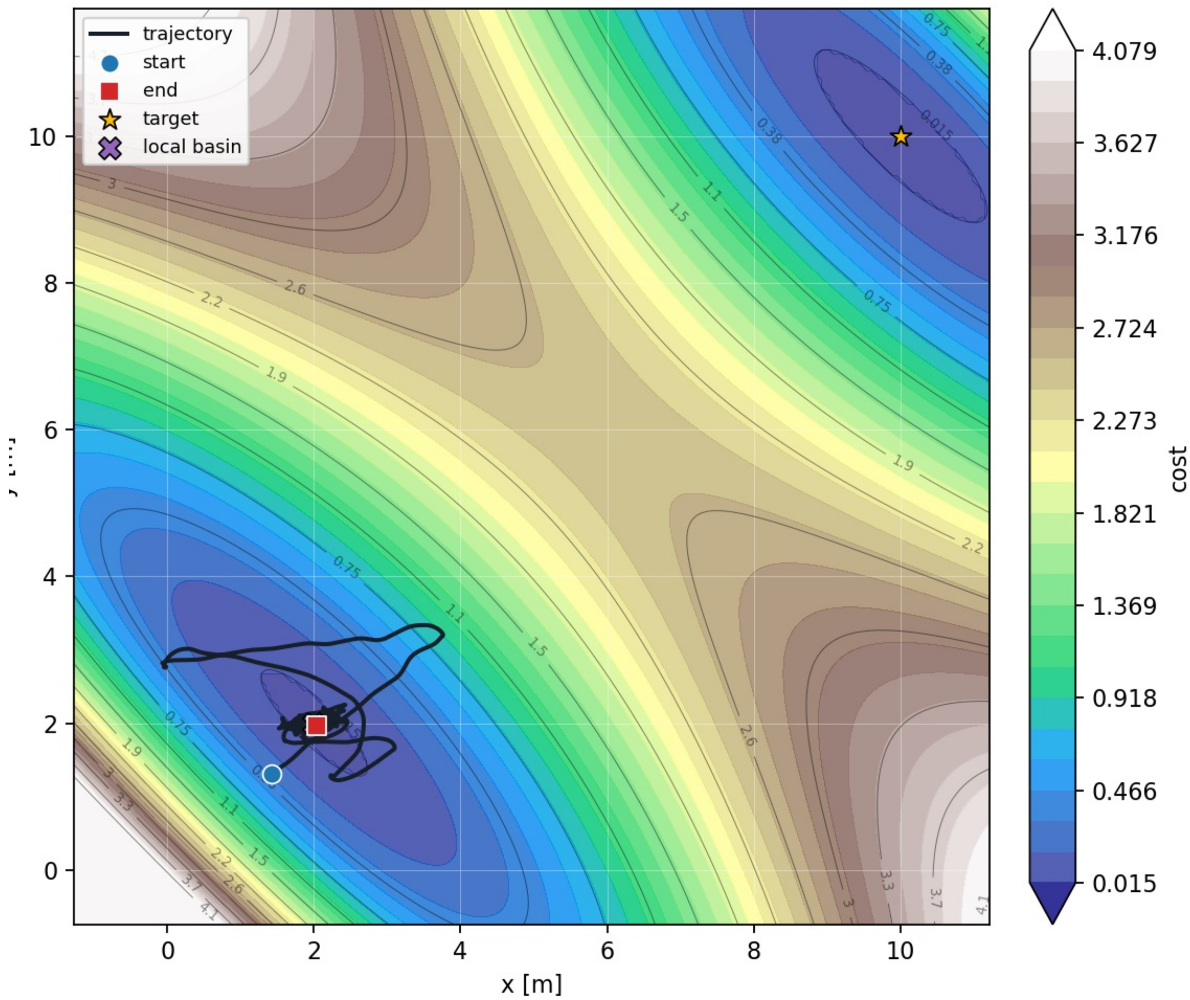

Figure 10. Baseline HBESC did not enter the target radius on the representative quartic double-well row.

Trajectory: double-well Gaussian fill does not rescue

P5B-DW-A0P01-GF; final distance 13.338 m, one fill.

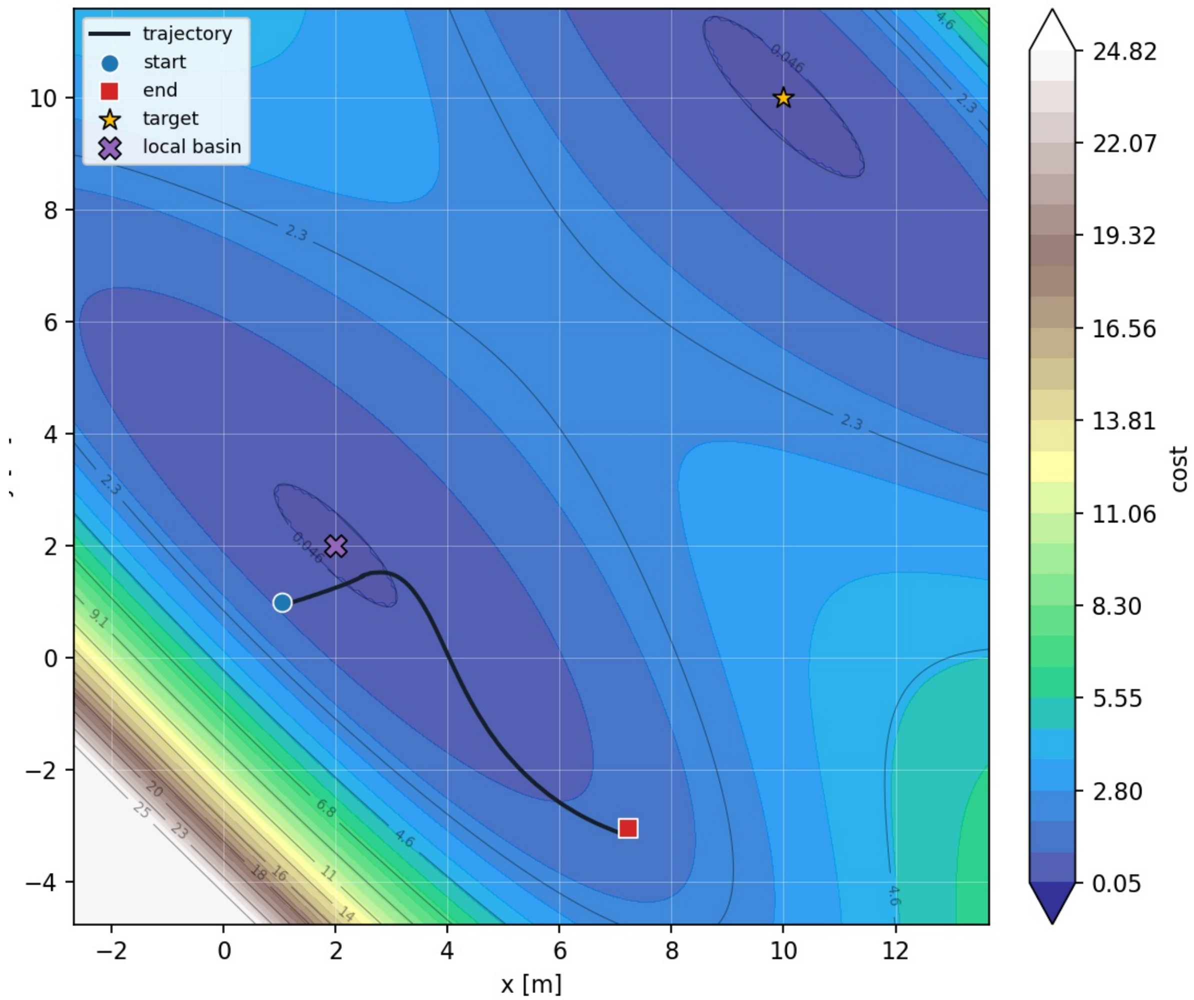

Figure 11. The paired one-fill Gaussian-fill row also failed, motivating targeted fill-placement diagnostics instead of another broad sweep.

Limitations And Next Step

This is a compact visual report, not the full raw-results appendix. The full audit remains in `results_manifest.csv`, per-run folders, and aggregate metrics CSVs.

Recommended next work: inspect Gaussian-fill placement and modified-cost diagnostics for the failed double-well rows, then run a small targeted fill-parameter matrix rather than another broad sweep.

Further Questions

- Where exactly were fills placed in failed double-well rows?
- Would larger amplitude, different sigma, earlier trigger, or multi-fill behavior change the double-well result?
- Which gain settings remain useful inside a hardware-feasible wheel-RPM envelope?